

Silicon-Native QKD Transmitter for Bidirectional Quantum/Classical Co-Existence in High-Speed PON

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Abstract We prove secure-key generation over passive optical network plants for co-existing C- / O-band quantum/classical channels utilizing an all-silicon QKD transmitter, involving a 2×16 power split while being able to tolerate bidirectional classical launch levels up to 0 dBm. ©2026 The Author(s)

Introduction

The ongoing roll-out of QKD [1, 2] to secure high-value assets builds primarily on bulky and costly 19" rack-sized hardware. Therefore, research heavily focuses on compact and cost effective photonic-integrated QKD solutions [3-6]. Additionally, the introduction of QKD in mass-user applications such as passive optical access networks (PON) raises major challenges: First, no significant cost burden should apply when piggybacking PON transceivers with QKD functions. Second, QKD should overcome the high optical budget inherent to split-based PON architectures [7]. Third, QKD should be robust to Raman noise generated by down- and upstream data signals [8]. Although earlier works have studied the applicability of QKD in PONs [9-11] with respect to specific aspects, its compatibility with the entirety of the aforementioned deployment challenges is yet to be demonstrated.

In this work, we investigate the suitability of a monolithic silicon QKD transceiver [6] in light of these three challenges. We experimentally show classical/quantum co-existence for high-speed (HS) PONs by demonstrating a positive secure-key rate for simultaneous down- and upstream transmission at 0 dBm over a 2×16 optical distribution network (ODN) split.

Silicon-Native QKD for (V)HS-PONs

Figure 1a presents the concept for integrating C-band QKD in a (V)HS-PON, where classical data traffic is allocated to the O-band [12]. We build on the asymmetric complexity distribution inherent to unidirectional discrete-variable QKD protocols such as BB84, where costly single-photon receivers can be pooled at a central office (CO) and shared among multiple users – each of which is equipped with simplified and cost-efficient QKD transmitters. Towards that, our all-silicon QKD implementation ensures marginal cost increase when appending QKD functionality to the optical network terminals (ONT): As shown in Fig. 1b, the use of a silicon-native low-light source at 1550-nm [6] omits the need for a dedicated C-band optical power supply for QKD, as it is otherwise required for the classical O-band PON transmitter sub-system. Given that the related hetero-integration is subject to considerable

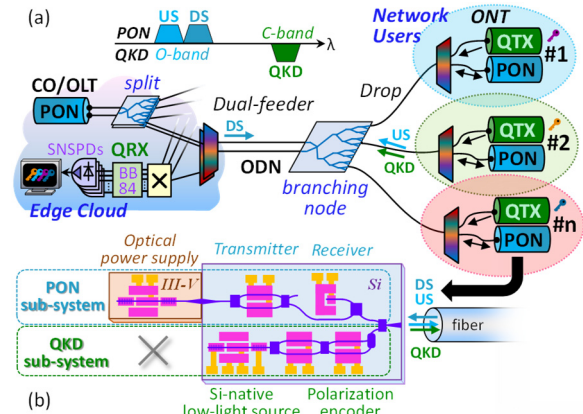


Fig. 1: (a) Integration of QKD in resilient (V)HS-PON architectures and (b) silicon-native QKD approach.

assembly and hermetic packaging costs [13], which can amount to a high share of 90% in the overall device costs, the addition of chip space devoted to QKD merely affects the silicon ONT platform, meaning an acceptable cost burden. We consider dual-homing PON architectures to further address co-existence aspects. These employ resilient feeders between the CO and the branching node within the ODN, followed by short drop spans of a few 100 m connecting the ONTs. Given the detrimental effect of Raman noise on the quantum signal [14, 15] caused by high-power classical signals over shared quantum/classical single-mode fiber (SMF) links, this layout can be advantageous: By introducing a dual-feeder architecture for separating quantum and classical signals with respect to downstream transmission, the fiber distance along which Raman noise is generated is greatly reduced, introducing resiliency at the shared trunk segment [16]. Furthermore, the wide spectral gap between (V)HS-PON in the O-band and the C-band quantum channel further reduces the impact of Raman noise.

Experimental PON for QKD Co-Existence

Figure 2a presents the experimental setup. The photonic QKD transmitter is realized on a silicon-on-insulator platform and wire-bonded to a PCB (Fig. 2b). The monolithic integrated SiGe light source is a forward-biased PIN junction, emitting broadband incoherent photons as described in detail in [6]. The four required polarization states

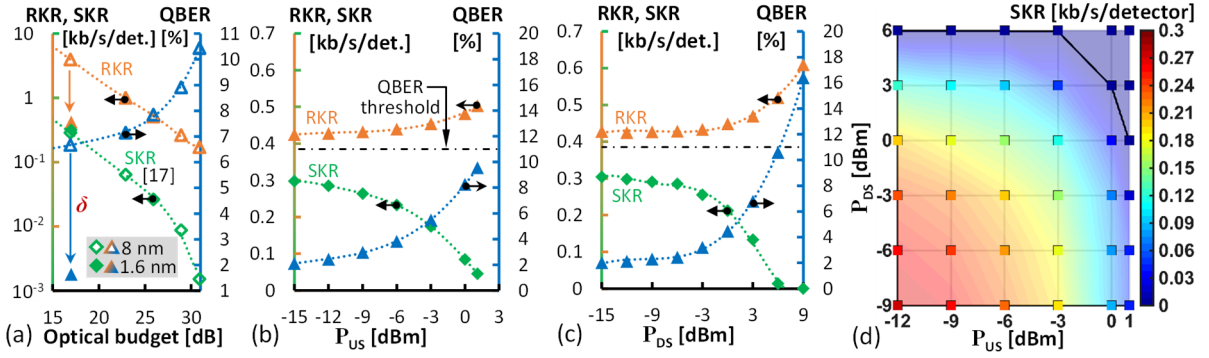


Fig. 4: QKD performance for (a) dark PON, as function of transmitter bandwidth, and for lit PON in case of (b) exclusive US launch at P_{US} , (c) exclusive DS launch at P_{DS} , and (d) bidirectional US and DS operation with simultaneously launched P_{US} , P_{DS} .

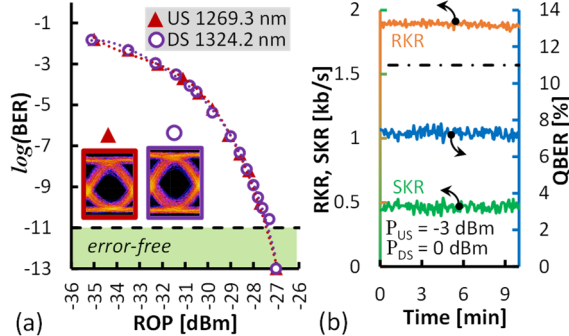


Fig. 5: (a) DS / US BER. (b) long-term QKD performance.

potential US wavelengths λ_{US} in the 1270 nm, 1290 nm and 1310 nm CWDM-bands has been investigated using a single-photon resolving optical spectrum analyzer. As Fig. 3c shows, λ_{US3} at 1310 nm still induces significant in-band noise of ~ 5.5 kcts/s/nm at the quantum channel λ_Q when setting an US launch power of 0 dBm. When re-allocating the US to λ_{US2} at 1290 nm, the noise at λ_Q reduces to ~ 1.5 kcts/s/nm due to the spectral roll-off in Raman noise at a detuning of ~ 260 nm from its sourcing signal. λ_{US1} at 1270 nm further improves the in-band noise as it leads to an almost flat Raman background at ~ 700 cts/s/nm in the C-band. This US wavelength, which is compatible with the spectral options of HS-PON [7], was therefore chosen for the following co-existence studies.

A previous back-to-back evaluation of the QKD transmitter in a dark ODN setup [17] yielded a compatible optical budget of 30.9 dB for an 8-nm wide QKD channel (Fig. 4a). The reduced 200 GHz (1.6 nm) bandwidth yields a lower QBER of 1.64% since the fiber-induced depolarization penalty is mitigated (δ). The estimated SKR [19] slightly improves from 288 to 318 b/s/detector. In addition, the smaller filter bandwidth offsets the impact of in-band Raman noise in a lit ODN.

Towards that, Fig. 4b shows the increase in RKR and QBER for the lit PON due to US-generated Raman noise at the drop fiber. For the highest launch of $P_{US} = 1.1$ dBm we could still generate a secure key at 46 b/s/detector. This is sufficient to secure 0.55 Tb/s/detector of PON traffic considering the AES-GCM limit for single-key use over TLS 1.3 [20]. Without the C/O-

waveband splitter stack the Raman noise that is predominantly generated by the US prevents secure-key generation at typical levels of 0 dBm.

For co-existence with only the DS channel higher launch levels P_{DS} of up to 6.2 dBm can be tolerated before the 11% QBER cut-off for secure key extraction [18] is reached (Fig. 4c). This increase in tolerable co-existence power is attributed to the high attenuation of the DS through the ODN splitter before it generates Raman noise along the drop segment. The reduced Raman noise impact is also confirmed by the delayed onset of the QBER rise at higher optical powers than in the US case in Fig. 4b.

Figure 4d shows the achievable SKR under simultaneous DS/US operation, with the SKR cut-off being indicated by the solid line. A maximum SKR of 272 b/s/detector can be accomplished at $P_{US} = -12$ dBm and $P_{DS} = -9$ dBm. For the maximum P_{US} of 1.1 dBm, a secure key can be generated at 20 b/s/detector for a P_{DS} of up to -3 dBm, while for a higher P_{DS} of 3 dBm the US needs to be levelled to $P_{US} = -3$ dBm for secure-key generation at 52 b/s/detector.

Furthermore, we investigated the classical channel performance, which shows a reception sensitivity of -27.4 dBm at a BER $< 10^{-11}$ given the symmetric DS/US transmitter layout (Fig. 5a). Due to the actual ODN end-to-end loss, this sensitivity yields a lower bound for the classical channel launch at -9.9 dBm and leaves a power margin of up to 9.9 dB before reaching the SKR cut-off due to classical / quantum co-existence penalties. Finally, Fig. 5b confirms stable SKR operation at 454 b/s under slightly relaxed conditions of $P_{US} = -3$ dBm and $P_{DS} = 0$ dBm.

Conclusions

We have verified secure-key generation employing a silicon-native QKD transmitter implementation in a (V)HS-PON scenario, taking advantage of the O-band-centric US options foreseen in HS-PON. This case study proves the potential of simplified all-silicon QKD transmitters in achieving positive SKRs in access applications involving residential end-users, allowing QKD to enter domains deemed as challenging so far.

Acknowledgements

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